IS 107 – Business Intelligence

Laboratory Work 4

Forecasting Time-Series Data in Power BI

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Forecasting | **Time-Series** | **Power BI** | **Analytics**

Lab 4 – Overview

Part 1: Tourism Data

- Download tourism-data.csv
- Import into Power BI
- Create Line Chart (Yearly)
- Add Trend Line
- Add Forecast (12 pts, 95% CI, Seasonality 12)
- Analyze forecast for 2020

Part 2: Imperfect Data

- Use tourism-data-modified.csv
- Dataset includes COVID disruption (2020–2021)
- 50 new data points added
- Experiment with forecast settings
- Analyze confidence intervals
- Compare Ignore Last settings

Analysis & Report

- Describe dataset & source
- Visualize forecast with screenshots
- Discuss trend & seasonality
- Interpret confidence intervals
- Discuss disruption handling
- Conclusions & limitations

P A R T 1

Tourism Data Forecast

Dataset: tourism-data.csv • Objective: Apply Forecasting in Power BI

Steps 1 & 2: Import Data & Create Line Chart

1 Download Dataset

Go to the GitHub link and save tourism-data.csv to your computer.

2 Import into Power BI

Home → Get Data → Text/CSV → select and Load tourism-data.csv

3 Create Line Chart

Drag Period to the X-axis, Total tourists to the Y-axis, then set hierarchy to Year + Month.

tourism-data.csv

File Origin: 1252: Western European (Windows)

Delimiter: Comma

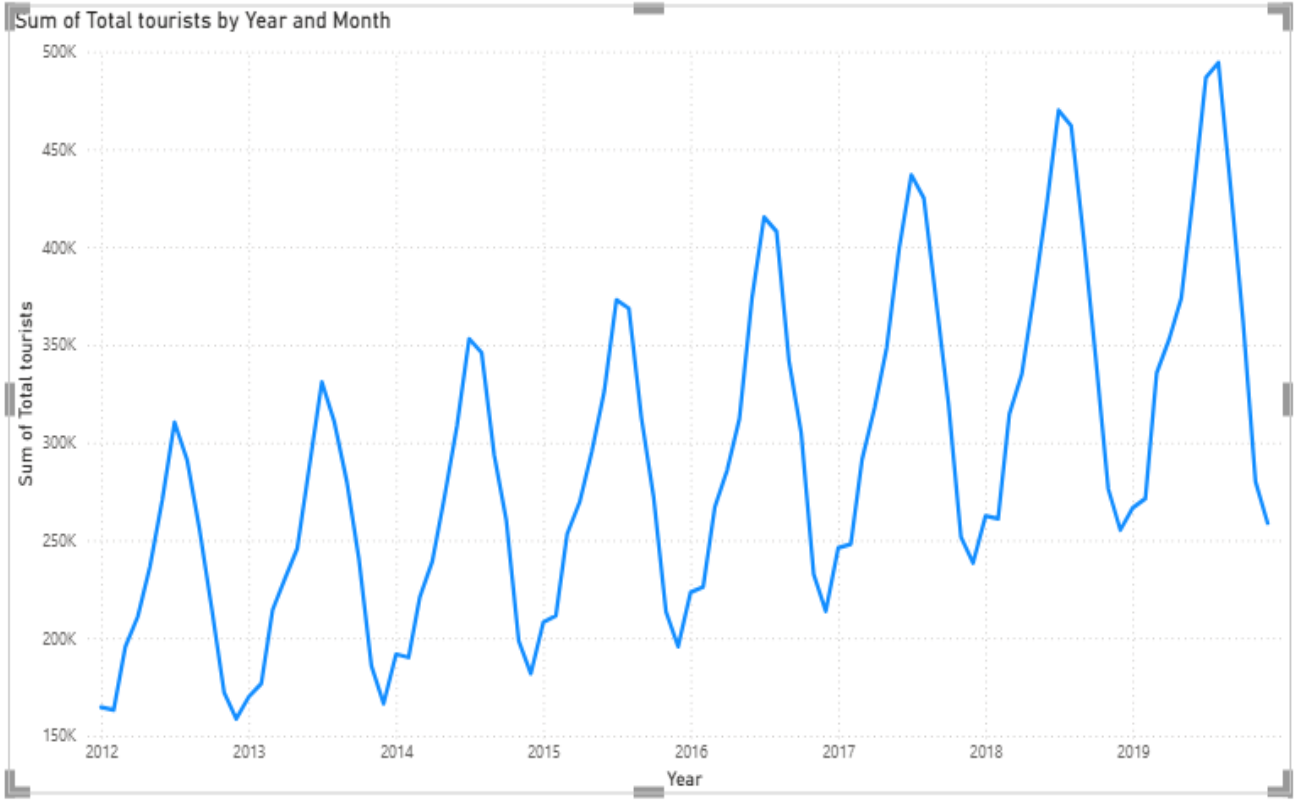
Data Type Detection: Based on first 200 rows

Period	Total tourists
1/1/2012	164699
2/1/2012	163212
3/1/2012	195656
4/1/2012	211101
5/1/2012	236643
6/1/2012	271178
7/1/2012	310471
8/1/2012	291264
9/1/2012	255464
10/1/2012	215412
11/1/2012	172045
12/1/2012	158717
1/1/2013	170028
2/1/2013	176747
3/1/2013	214411
4/1/2013	230620
5/1/2013	245806
6/1/2013	289039
7/1/2013	331033
8/1/2013	310524

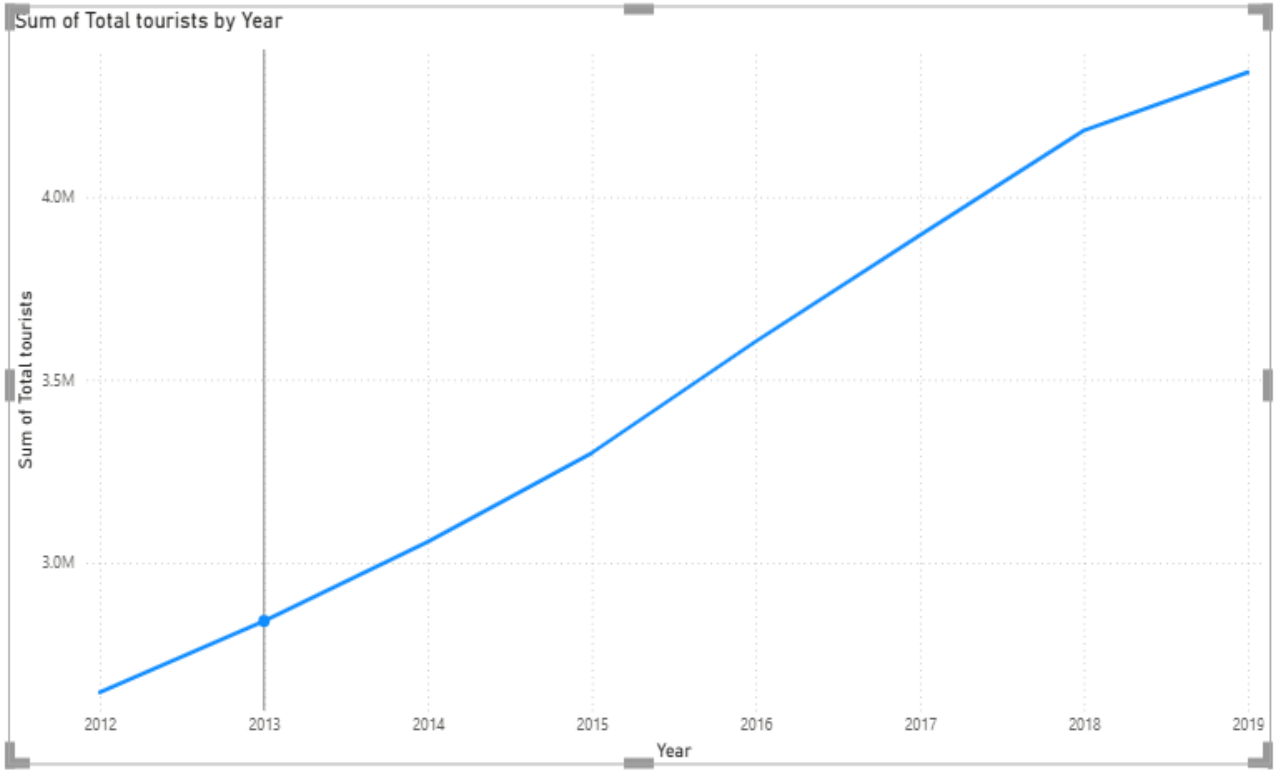
The data in the preview has been truncated due to size limits.

Extract Table Using Examples | Load | Transform Data | Cancel

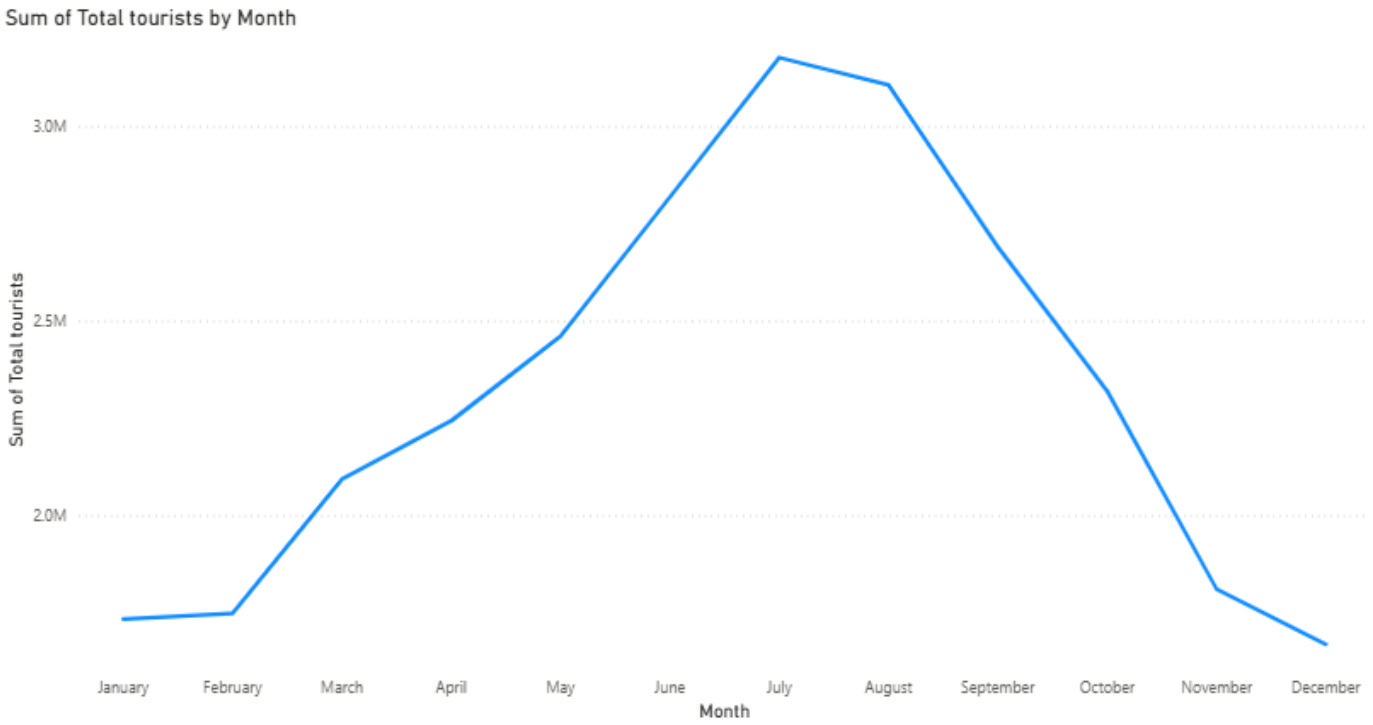
Steps 1–3: Create Line Chart by Year + Month



Steps 1–3: Create Line Chart by Year



Steps 1–3: Create Line Chart by Month



Step 4: Add a Trend Line

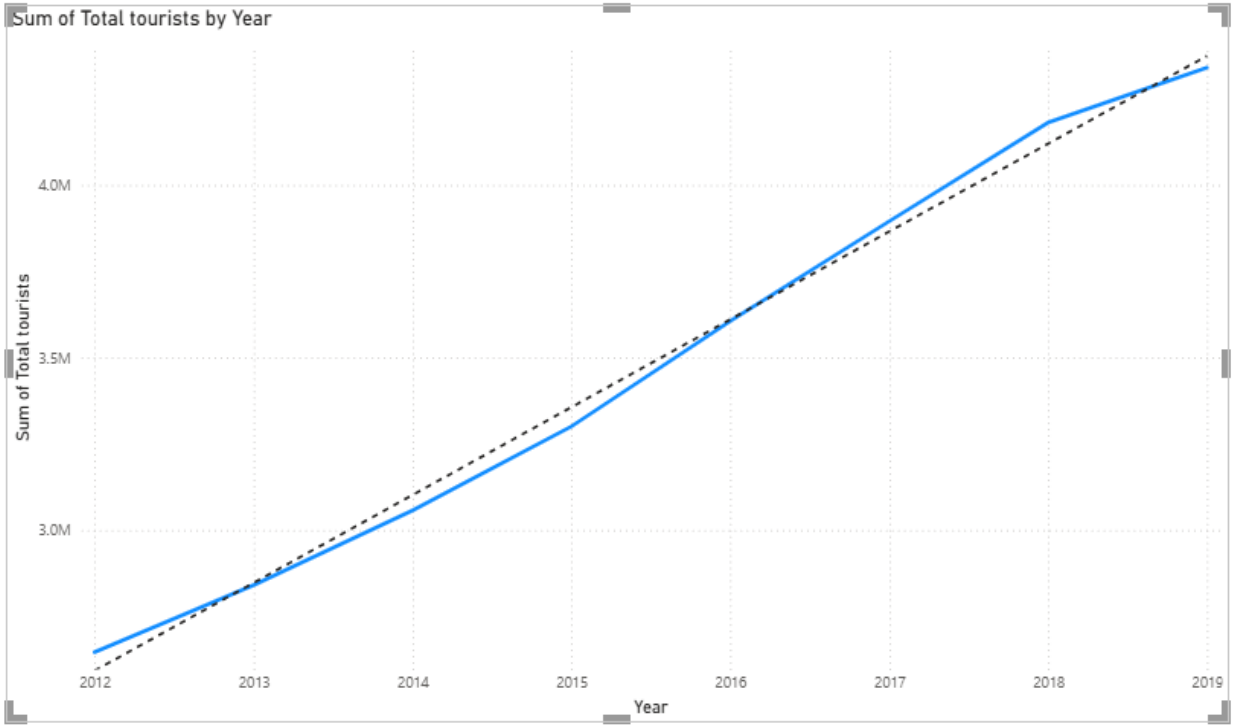
1 How to Add the Trend Line

- Select the line chart on your canvas
- Open the Analytics pane (magnifying glass icon)
- Expand the 'Trend line' section
- Click + Add to apply

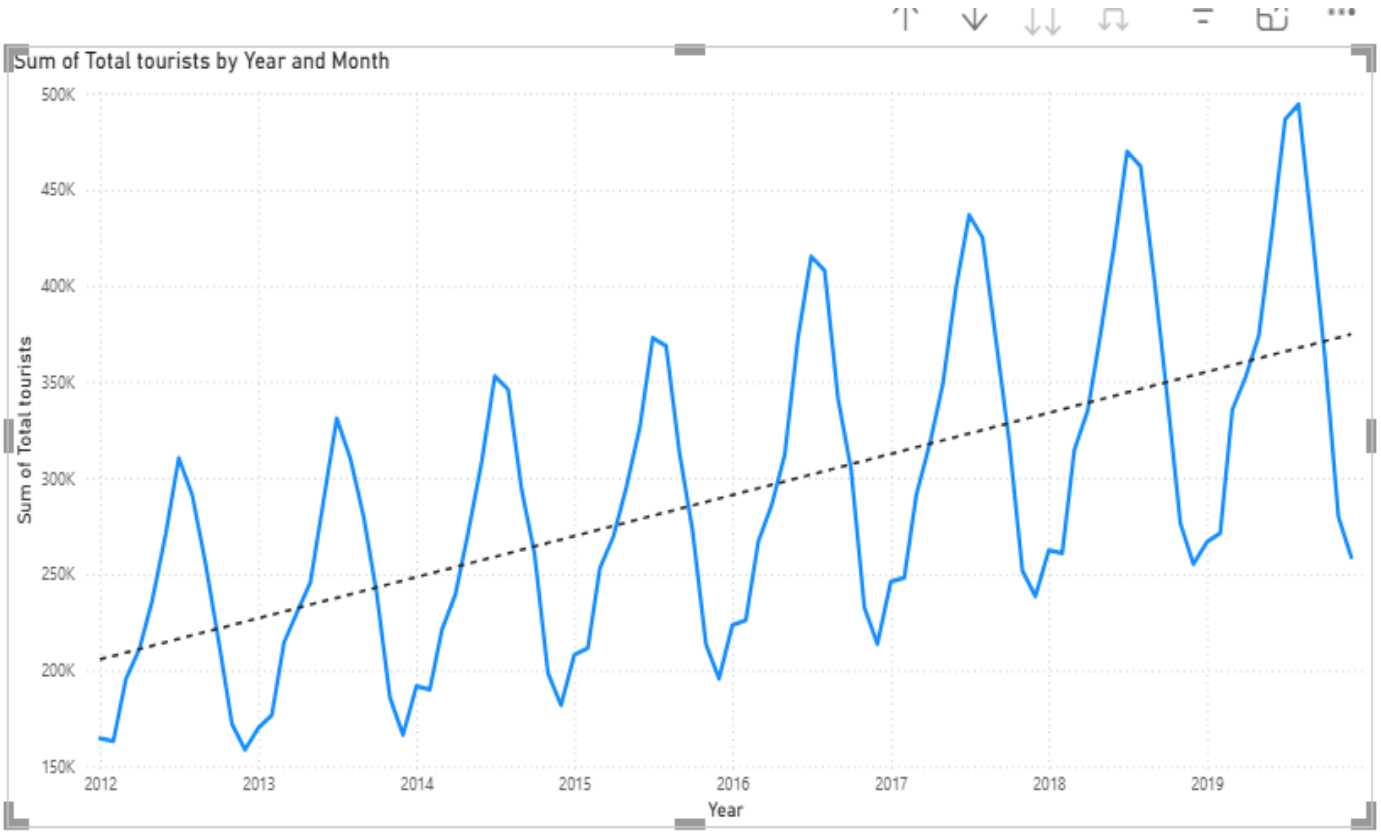
2 What you should see

A dashed diagonal line overlays the chart, reflecting a consistent upward trend in visitor numbers from 2012 through 2019.

Step 4: Add a Trend Line By YEAR



Step 4: Add a Trend Line By Month and Year



Step 5: Add Forecasting

Forecast length

10 points

Confidence interval

95%

Ignore last

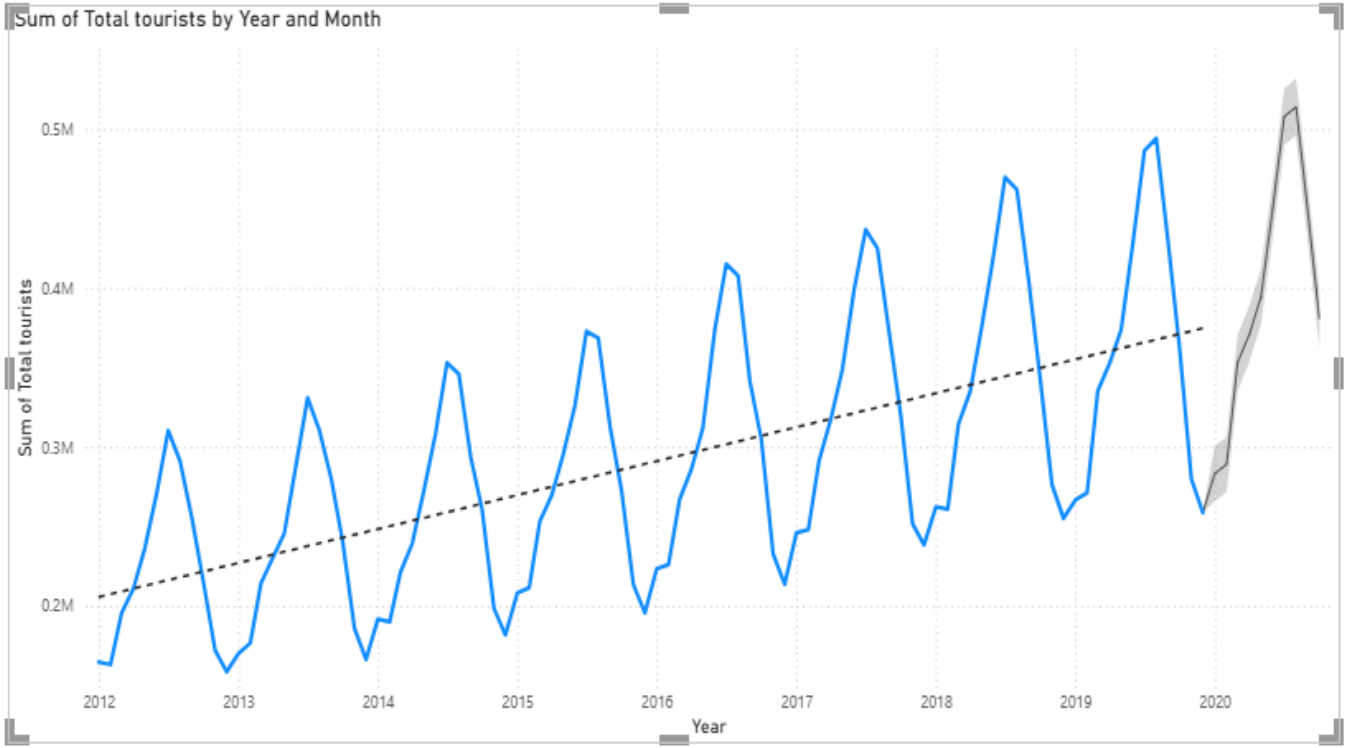
0 points

Seasonality

...

After clicking Apply, the forecast extends beyond 2019 as a shaded band representing the 95% confidence interval.

Step 5: Add Forecasting



Step 6: Configure the Forecast

Forecast length

12 points

Confidence interval

95%

Ignore last

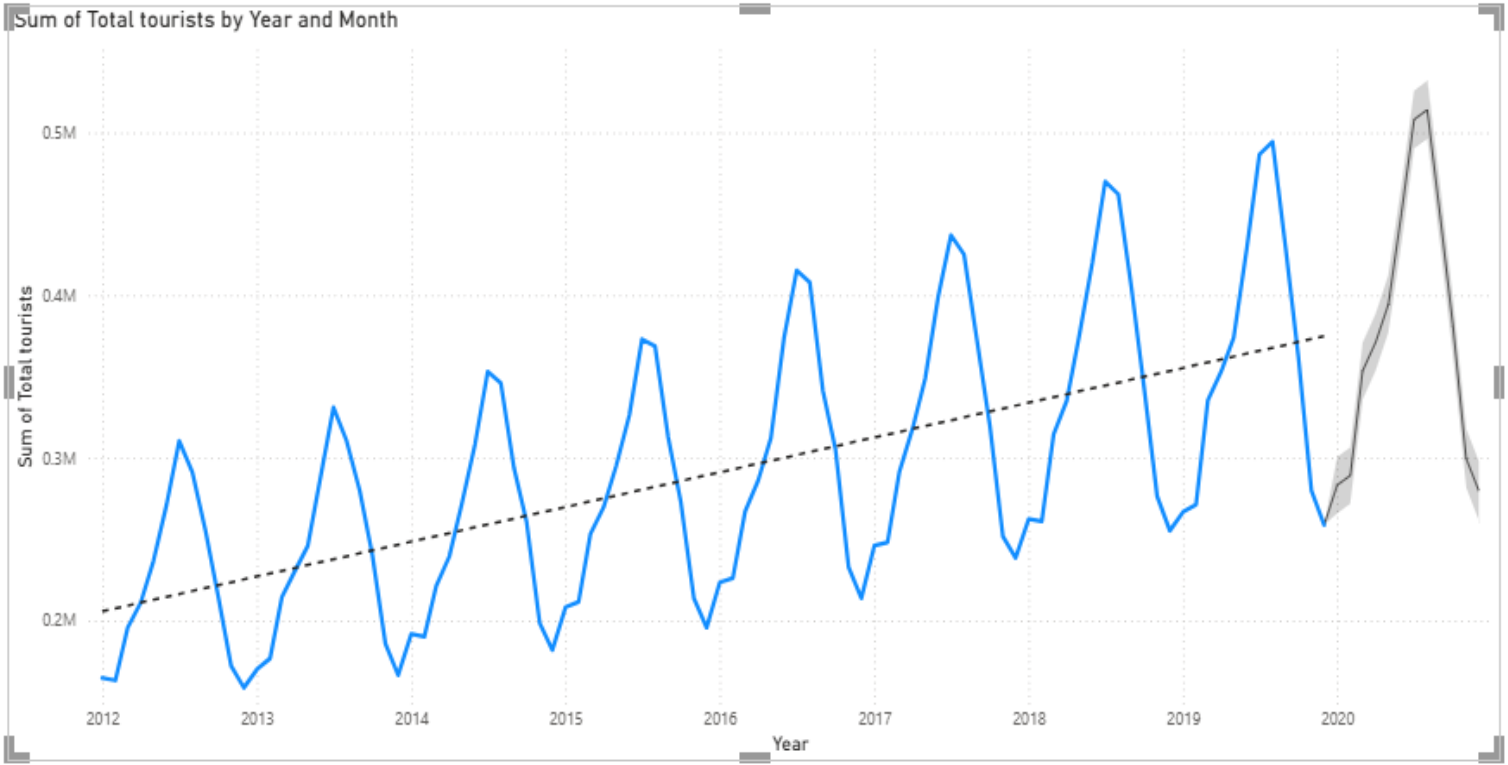
0 points

Seasonality

12 points

After clicking Apply, the forecast extends beyond 2019 as a shaded band representing the 95% confidence interval.

Step 5: Add Forecasting



Step 7: Analyze the Forecast

1 Trend

Tourism numbers rose steadily from roughly 2.8M in 2012 to 4.6M by 2019, and the model captures this year-over-year growth trajectory.

2 Seasonality

Visitor peaks occur each July–August while January–February consistently see the lowest counts. Setting Seasonality = 12 correctly identifies this annual pattern.

3 Confidence Interval

The 95% CI band appears relatively narrow, signaling strong model confidence. Stable, recurring patterns in the training data yield reliable future projections.

P A R T 1

Tourism Data Forecast

Dataset: tourism-data.csv • Objective: Apply Forecasting in Power BI

Part 2 – Imperfect Data Overview

Dataset: tourism-data-modified.csv

- Spans Jan 2012 – Feb 2024 (146 rows total)
- 50 data points appended beyond the original dataset
- COVID-19 impact simulated across 2020–2021
- Recovery trajectory continues through 2022–2024

Observed Phases

2012–2019	Steady growth with consistent seasonal waves
2020	Near-collapse driven by pandemic lockdowns
2021	Gradual partial recovery begins
2022–2024	Strong rebound toward pre-pandemic levels

Experiment 1: Baseline Forecast

Forecast length

12

Seasonality

12

Ignore last

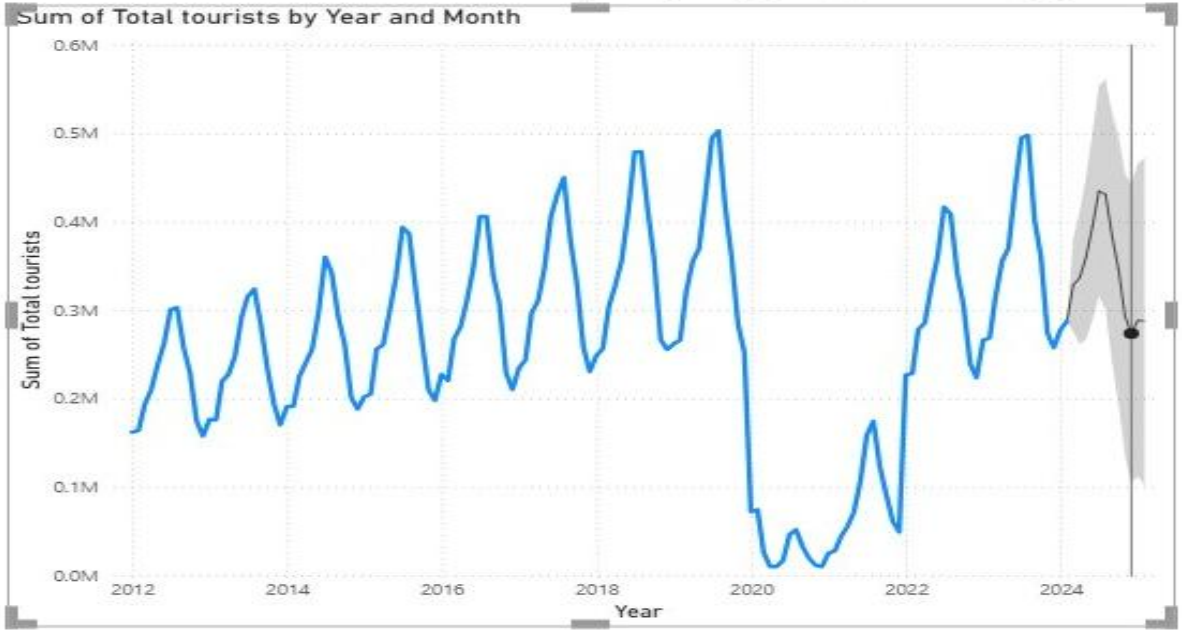
0

Confidence interval

95%

The forecast carries forward the recovery momentum. The CI width is moderate — COVID disruption in the training data increases variability, yet the model still projects gradual seasonal growth.

Experiment 1: Baseline Forecast



Experiment 2: Ignore Last 6 Points

Forecast length

12

Seasonality

12

Ignore last

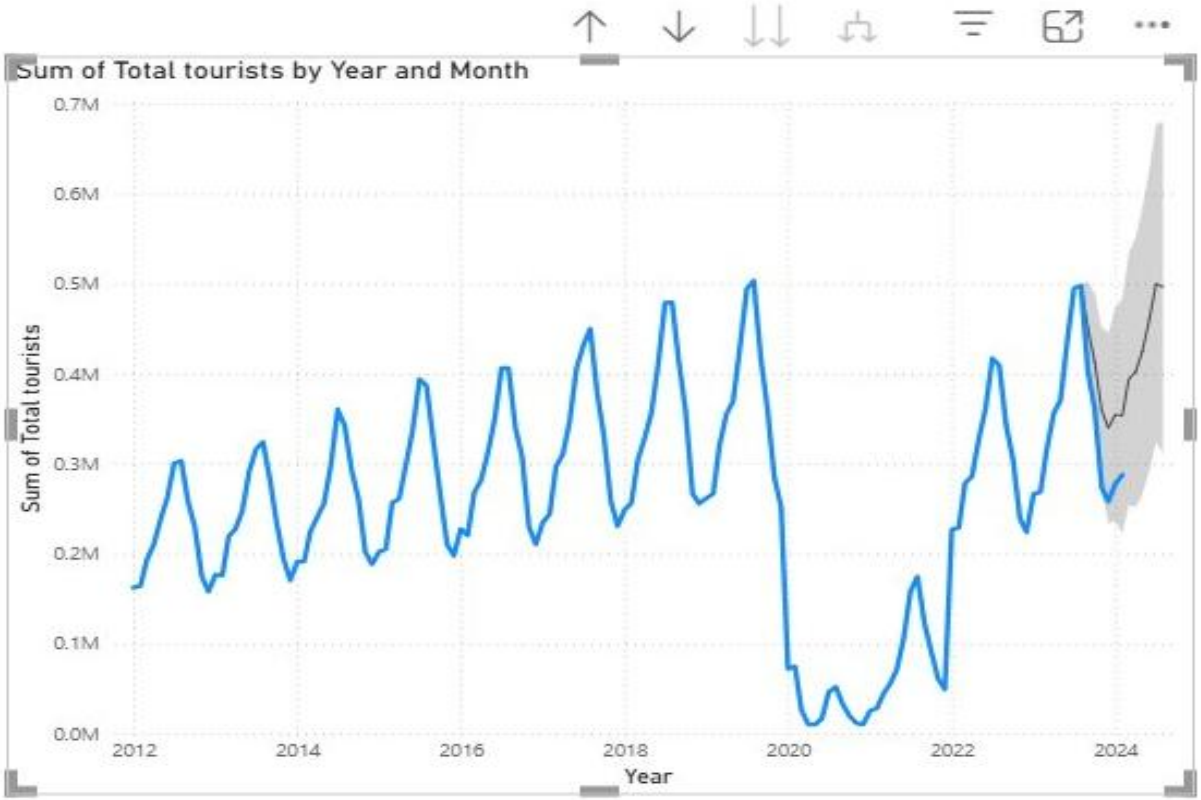
6 ← changed

Confidence interval

95%

Dropping the 6 most recent months moves the forecast start point backward. The CI broadens slightly, since the model now has less up-to-date evidence — introducing more uncertainty.

Experiment 2: Ignore Last 6 Points



Experiment 3: Higher Confidence Interval

Forecast length

12

Seasonality

12

Ignore last

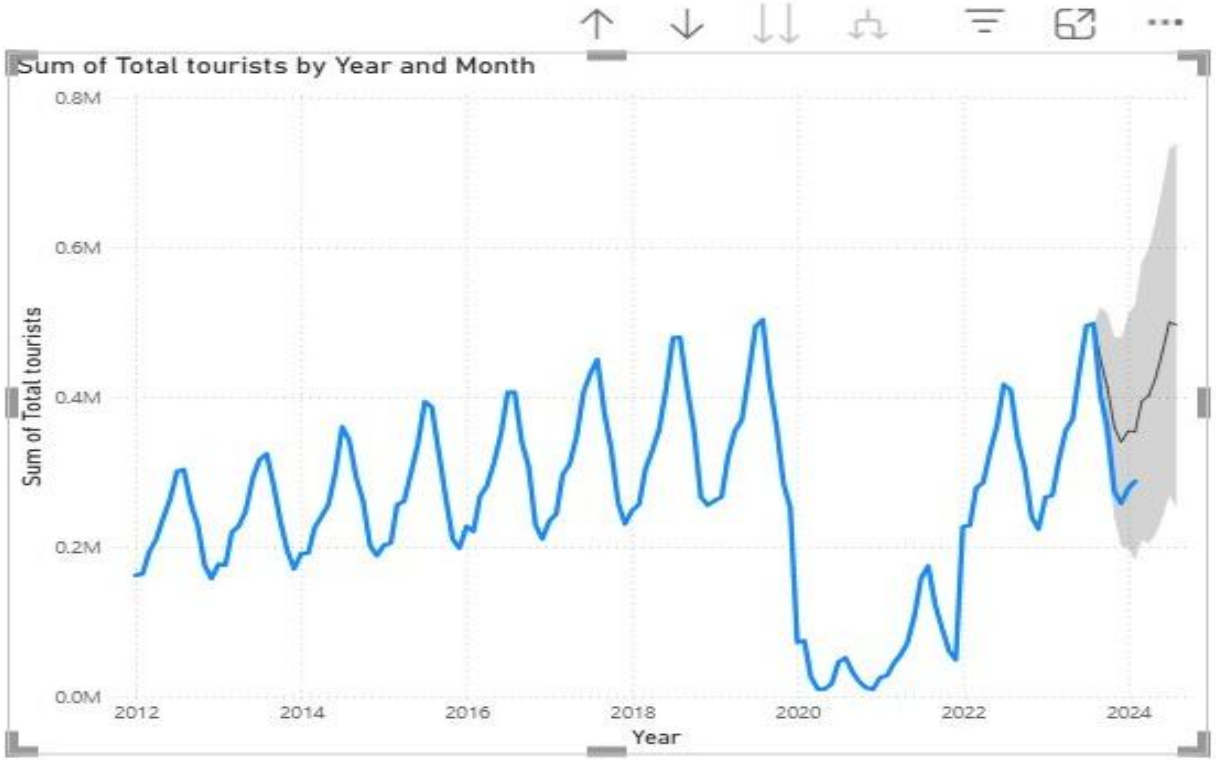
6

Confidence interval

99% ← changed

Raising the CI to 99% noticeably widens the shaded band. Achieving greater certainty forces the model to span a larger possible range, highlighting the elevated uncertainty left by the COVID shock.

Experiment 3: Higher Confidence Interval



How the Forecast Handled the COVID Disruption

Did NOT Predict the Disruption

Power BI has no mechanism to anticipate sudden external events like a global pandemic from historical data alone — the abrupt crash was beyond the model's scope.

Incorporated Recovery Trend

Post-disruption figures from 2022–2024 pulled the forecast upward, causing the model to project continued growth in line with the observed rebound.

Wider Confidence Intervals

All three experiments produced broader CI bands than Part 1 — a direct consequence of the disruption inflating variability within the training dataset.

Seasonal Pattern Disrupted

The irregular 2020–2021 period broke the regular annual cycle, weakening the model's ability to estimate seasonal patterns with confidence.

Guide Questions & Answers

What does a narrow confidence interval suggest?

A tight CI means the model is operating with high confidence. When training data follows a stable, repeating pattern, the algorithm can project a smaller range of likely outcomes.

How does raising CI from 95% to 99% change the visual?

The shaded region becomes considerably wider. A higher CI demands that the model cover a broader band of possible outcomes to meet the stricter certainty requirement, making the uncertainty more visible.

How does the Ignore Last setting affect the forecast?

Configuring Ignore Last = 6 removes the six most recent observations from the model's training window. This shifts the forecast start backward and slightly widens the CI, reflecting greater uncertainty from reduced data recency.

How useful is Power BI forecasting for this data?

Very useful when the data is clean and consistent (Part 1). With disrupted data (Part 2), the widening CI limits actionable precision — situations like these may call for more sophisticated statistical or ML approaches.